

SmartGuard AI

Edge AI Based Driver Safety System

1. Project Overview

SmartGuard AI is a proposed real-time driver safety system that combines Edge AI, computer vision, and sensors to detect unsafe driving conditions and provide immediate alerts. The system is designed to process camera information locally on an edge device, reducing dependence on cloud processing and improving response time and privacy.

2. Problem It Solves

Traditional vehicle safety systems may depend on separate sensors, cloud services, or delayed analysis. SmartGuard AI aims to provide fast local detection of driver and road-safety conditions and generate appropriate alerts in real time.

3. System Architecture

Input	Processing	Output
Camera	Edge AI / Computer Vision	Driver Safety Alert
MQ-3 Sensor	Safety Decision Logic	Buzzer / OLED
GPS Module	Emergency Handling	Location Alert

4. Tools & Technologies

- Python
- OpenCV
- YOLO / Computer Vision
- Raspberry Pi
- Camera
- MQ-3 Alcohol Sensor
- GPS Module
- Buzzer and OLED Display
- GitHub

5. My Contribution

Designed the system architecture and developed the Edge AI concept for local camera processing. Planned the integration of sensor inputs with the AI-based safety decision system and designed the real-time driver alert and GPS-based emergency notification workflow.

6. Key Features

- Local Edge AI processing

- Real-time camera-based detection
- Sensor-assisted driver safety monitoring
- Immediate warning alerts
- GPS-based emergency notification concept
- Reduced dependence on cloud processing

7. Future Enhancements

Future versions can include drowsiness detection, phone-use detection, improved object detection, optimized AI models, cybersecurity features, and a fully integrated vehicle safety prototype.

Project Type: ECE / Embedded Systems / Edge AI